**Insulin Pricing: Correcting Market Failures**

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## Introduction

Insulin prices continue to skyrocket, uncapped, leaving Americans with diabetes to face high, stressful rates. There are many factors that have contributed to the rising prices of insulin in the United States such as the manufacturers themselves and price hikes at pharmacies, allowing the customers to be deceived. These indicate a market failure in perfect information has occurred, patients are given prescriptions and may not know where they can go to get the best price for their medication. This project aims to lessen the gap between the manufacturers and consumers, so consumers of insulin have more insight into pricing, and where they can go for the prices that are best for them. To do so, the goal of this project is to create a website where consumers of insulin can go to compare prices and get information about the high cost of their medication.

## Literature Review

Professors Edwin Gale and John Yudkin discuss politicalization of insulin in their commentary in the British Medical Journal piece titled "Politics of Affordable Insulin". They begin their commentary discussing how there has yet to be a generic brand of insulin to hit the market, even though human insulin is no longer on a patent. Generic biosynthetic insulin is not being produced even though many companies have the equipment to produce the lifesaving generic alternative to name brand insulin, yet they are not doing so. Gale and Yudkin claim that this is because of the marketing and narrative around brand name insulin being better acting or more effective. According to Gale and Yudkin, participants in a study done, "were unable to tell the insulins apart at the end of the study and expressed no clear preference either way" (2011, p. 566). Participants in their study were unable to tell a difference between branded and generic insulin, yet the companies selling branded insulin sell it at a markup of 200-400% according to the researchers. After doing research Gale and Yudkin ask themselves why there is such a markup on insulin and find their answer in big pharma business, and the market power of three companies.

There are three companies that control the market for insulin, creating an oligopoly in insulin manufacturing. The professors reference a study done in which volunteers from 60 countries go to pharmacies to purchase ten milliliters of human insulin, Gale and Yudkin (2011) found that, “The price varied by a staggering 5000%, and by 2900% for the same brand” (p. 566). This outrageous difference in price for the same product is due to what Gale and Yudkin describe as the pharmacies and suppliers taking a large profit from the insulin that they are selling. Access remains a large problem for many families and persons struggling with diabetes and thousands of people die due to not being able to afford insulin. Gale and Yudkin claim that the companies cannot be blamed for wanting to make a profit off of the insulin they are selling, companies want to continue to develop products, and sell at prices that can be supported by the markets for insulin, which is strikingly inelastic, and avoid cuts in prices. Gale and Yudkin call for competition and open markers between generic and branded insulin to fix the price gouging or at least slow down the absurd price consumers pay for insulin. Of course there are challenges facing generic insulin companies, including the base costs of creating insulin, and being outmarketed by the three big companies.

The process of insulin getting to the point that it is now has been tumultuous, for all insulin companies. Now, with prices skyrocketing and a lack of competition between companies this article shows failures in the market that have occurred with insulin. The three big companies that have created the oligopoly are Eli Lilly, Novo Nordisk, and Sanofi. They push generic competition out of the market. They do this by having access to facilities that generic competitors don't have as well as marketing that guides consumers away from using generic alternatives. This article also goes into the lack of perfect information, the price of insulin changes from pharmacy to pharmacy even for the same brand of insulin, consumers do not always know where they could get the cheapest form of their prescription or human insulin especially because the price could change from store to store. This is why our capstone project is so important, we are attempting to correct the imperfect information in the market so that consumers can see what different prices their insulin could be in different places.

Deborah Cohen and Philip Carter agree with Yudkin and Gale that insulin has become unaffordable for the average consumer. Both sets of writers discuss the outrageous pricing not being properly communicated to consumers. Cohen and Carter agree with Yudkin and Gale, their article stating that part of the reason that costs are so high is the manufacturers of insulin, going into those details further, going through how much money manufacturers are reaping from outrageous prices.

Cohen and Carter in their article "How Small Changes Led to Big Profits for Insulin Manufacturers" (2011) explain pricing of insulin, and how large profits for the manufacturers came to be. For people with diabetes, the extra cost of insulin analogs is often not worth the benefit; it costs twice as much. The market is dominated by the companies that are pushing this new more expensive kind of insulin even though it is not as effective. Analog insulin is not any more effective, or any harder to produce,  making the high price of analog insulin confusing. Cohen and Carter suggest that the high market share comes from the amount of marketing done by the big insulin companies, including clever tactics that made the analog insulin appear more successful and safer than it is. The argument that Cohan and Carter make about the role of marketing in insulin pricing is similar to Gale and Yudkin's argument. Both sets of researchers agree that the marketing tactics of the big insulin companies have further alienated generic brands in the market. The point of our research and deliverable is to communicate this failure of perfect information in the market, to communicate data clearly and effectively, and help those with diabetes find where the cheapest type of insulin for them can be obtained.

Much like what Cohen and Carter mentioned in their article, Crozier-Fitzgerald attributed some of insulin's high price to the pharmacies, taking extra profit on insulin by taking advantage of the medication's inelastic demand. In her article "The Benefit is in the Details…" Crozier-Fitzgerald argues that patients should be able to access the medications that allow them to live while not spending all of their money. Crozier-Fitzgerald explains that in the United States, "9.3 million patients use insulin to manage their condition. Between 2002 and 2013 the cost of insulin tripled, then doubled in 2016, and rose again by about 5% in 2019. The price of insulin has become so high that one in four diabetic patients have reported insulin underuse because they cannot afford to buy their prescribed dosage" (2023, p. 97). These high prices are what are drawing in economic researchers to figure out what happened, specifically what market failures could be pointed to.

Crozier Fitzgerald uses the idea that the high cost of insulin can be attributed to a market failure, a lack of competition as well as the lack of competition among the main three Business Process Management (BPMs) companies that negotiate the price of insulin for the consumer. She explains that, the three main components of a drug's price, according to pharmaceutical companies, are the cost of development, free market economy forces, and continued innovation to compete with other companies; "1) insulin is 100+ years old so development costs are basically 0; 2) there are no free market forces present; 3) "innovation and competition among companies is meant to drive prices down for the benefit of the consumer, not to drive prices up for the benefit of the producer"(p. 104-5) There have been many policy proposals by the Trump and Biden administrations. Crozier-Fitzgerald would be in agreement with Gale, Yudkin, Cohen, and Carter, a market failure in information has occurred.

In contrast, researcher Schmitz argues that policy proposals should not consider price controls, which Crozier-Fitzgerald considers for the life saving medication, as they lead to deadweight loss. Schmitz explains that price controls are an issue as they cap a price where consumers are willing to pay more for a product, giving way to a deadweight loss and stop the market from responding to the demand that stemmed from the high prices. The article includes a counter argument, some may argue for price controls out of a moral place instead of following the market and efficiency. This, however, goes back to who wins and loses when it comes to price controls. Schmitz argues that the market is a language, and allows expectations to be formed. His argument is Neoclassical in nature, that the market allows prices to be set naturally at the best rate for the producers and consumers, clearing at equilibrium.

When the market is clearing at the equilibrium price, every customer who wants a product, such as insulin, at that price is able to get it. By contrast, the problem, according to Schmitz, is that "when supply is truncated by an artificially low price ceiling, there is unmet demand" (2015, p. 228) He argues that if a good is scarce, the buyers who need it more will outbid those who don't, he uses the example of ice. If there is a scarcity of ice then the people who just want ice to chill their beer will desire it less than those who need it to chill insulin or baby formula. His conclusion asserts, "If we extend this concern to its logical conclusion, we cap prices at $0, because for any price above zero, we can imagine a desperate would-be consumer unable to afford that price. Capping prices at zero, however, ensures that no consumer has more than a fair share by ensuring that no one has any share at all" (p. 231) In his conclusion Schmitz asserts that as long as the market is not interfered with, all market failures will be corrected, and no one will go without their medication.

This project asks if there is a way to correct the market failure of imperfect information. Consumers are unable to tell where they can get the best price for their insulin due to the oligopody between the insulin manufacturing companies and the BPMs. A price cap will lower the amount of stock that can be produced, and not go along with the needs of the consumers, but there has to be another way. Companies must be held accountable for the prices they are pushing onto consumers, and the price differences between pharmacies and different companies must be known to fix the market failure that is currently occurring in the insulin market.

## Economic Theory

### Neoclassical:

Neoclassical economics analyzes the distribution of resources, goods, and services through the medium of the market. In a perfectly competitive neoclassical market, it is assumed that there are many small firms producing nearly identical products with low barriers to enter production. Meanwhile, both producers and consumers are expected to have near perfect information which informs their rational decision making regarding how much to produce and how much to consume. With these criteria satisfied, each individual actor's self-interest, through the "invisible hand" of the market, efficiently distributes goods and services.

The United States insulin market exhibits a market failure defined by the market mechanism failing to optimally distribute the socially optimal amount of insulin. From a neoclassical perspective, the failure is related to competitive market assumptions not being fulfilled. Insulin production in the United States is dominated by three firms, Eli and Lilly Company, Novo Nordisk Inc., and Sanofi U.S., that together produce more vials of insulin than all other American firms combined (U.S. Import Data, 2025). This market configuration is indicative of a production structure much closer to oligopoly, or few large firms controlling the market, than the assumed plethora of firms competing.

A second issue arises with a lack of perfect information in the market. Insulin is frequently marketed and prescribed as a brand name drug for a higher price than generic alternatives. While this could be interpreted as a failure of rationality (consumers would be paying more for a brand name product as posed to an identical generic one), it is more likely to be an informational problem in part caused by advertising campaigns which intended to bury the existence of generic alternatives (Crozier-Fitzgerald, 2023, p. 106).

The market failure in the production of insulin is further accentuated by insulin's primary use case. Given that insulin is a life saving drug that is used in order to ward off preventable death from diabetes, demand is effectively completely inelastic. Put simply, regardless of the price of insulin, consumers will always need the same amount. Given that even in storage insulin does expire, there is limited applicability in "stocking up" while the price is low. Inversely, since going without insulin essentially risking death, it is unrealistic to cut back on consumption while the price is high.

This inelasticity also hints to another concept that is relevant for understanding market dynamics of insulin: (positive) externalities. While consumption of insulin has benefits to the consumer as they are kept alive, the benefits of consumption are not limited to the individual consumer. Since individuals develop skills and connections that provide value to society throughout their lives, keeping an individual alive when they would have died constitutes a positive externality. The individual receives utility to themselves by being kept alive and utility is distributed to society through the ways that individual produces value in the world.

As a result of these two factors, a traditional supply and demand graph does not fit the insulin market. Whereas supply is typically downward sloping, in the insulin market, demand is effectively vertical meaning that consumers will demand the same amount of insulin regardless of the price level (see figure 1). In turn, this allows firms to increase the price of insulin and directly gain additional profits since consumers cannot go without or buy a cheaper alternative. Graphically this increased profit, described as producer surplus, is represented by the area between the supply curve and Q' (equilibrium quantity demanded) being greater in the inelastic market.

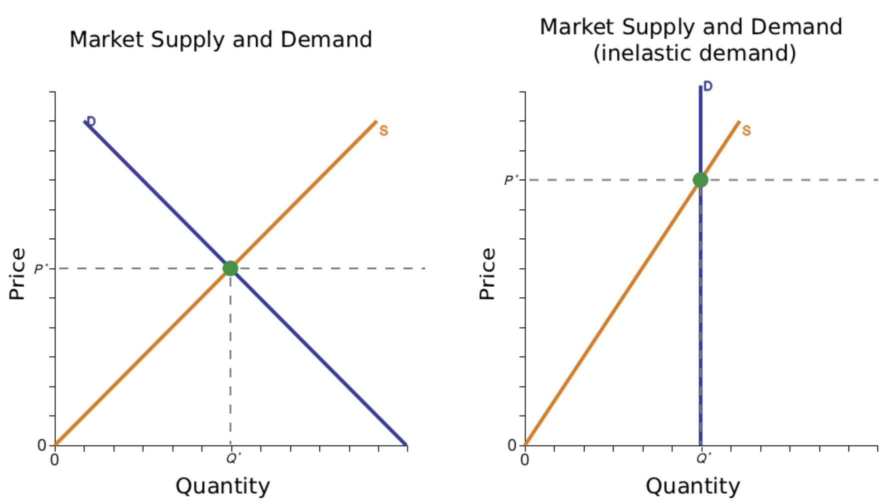


Figure 1

Furthermore, since there is additional social marginal benefit from the positive externality, the market equilibrium price of insulin does not produce a socially optimal quantity.

Graphically, while the utility consumption of insulin might create for the individual is represented by D, the value consumption of insulin might create to society is represented by D'. Since there is a gap between the optimal individual supply of insulin and the societally optimal supply, the area between the gap above the supply curve represents deadweight loss, or lost value to society (see figure 2). Should insulin reach its socially optimal supply (the intersection of D' and s), all the additional utility (represented as green) would be gained by society.

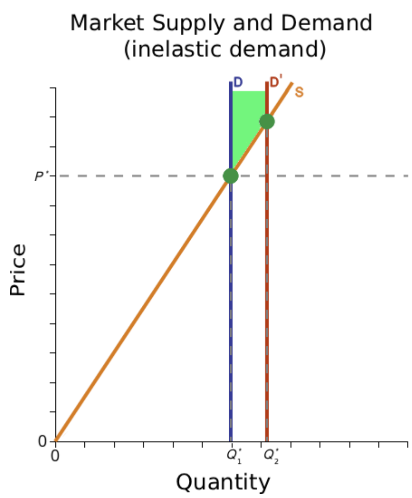


Figure 2

This discrepancy between the market supply of insulin and the socially optimal demand of insulin is compounded by the aforementioned lack of competitiveness of the insulin market. Lack of competition and suboptimal information has enabled the insulin manufacturing giants to raise prices over sixfold since 2002, raising insulin out of a comfortable price range for many Americans who have an inelastic demand for it (Crozier-Fitzgerald, 2023, p. 106). Whereas in a perfectly competitive market raising prices would cause consumers to go buy insulin from another producer, this market consolidation limits the availability of alternatives. This problem is amplified by insurance bureaucracies which, despite their purpose being insulin more accessible, create circumstances where generic alternatives are less common or not covered (Crozier-Fitzgerald, 2023, p. 107). Consequently, Americans who are unable to afford insulin at its oligopolized price could create deadweight loss between the marginal individual benefit and the marginal societal benefit of consumption while also potentially dying entirely preventable deaths. Conclusively, the failure of the market to properly supply insulin to all those who demand it is neither an issue of supply or demand, but rather of distribution being bottlenecked by pharmaceutical companies that distort supply and demand by abusing the lack of competition and information.

**Radical Political Economy**

A radical political economy (RPE) perspective identifies a lack of competition in the insulin market in a similar way. A lack of competition allows the oligopolistic firms to raise prices significantly past the costs of production, abusing the inelastic nature of insulin demand in order to generate maximal profits. Both neoclassical and RPE schools of thought believe firms to be profit maximizing, however, from an RPE perspective, the method of doing so is not in innovating new technologies that give a firm a temporary advantage, but rather in eliminating as much competition as possible. To this extent, RPE theorists like Michael Parenti claim that the premise of the invisible hand is idealistic and that it is unreasonable to expect, “that the most ruthless, selfish, opportunistic, greedy, calculating plunderers applying the most heartless measures in coldblooded pursuit of corporate interest and wealth accumulation will produce the best results for all of us” (Parenti, 2012). Whereas neoclassical economics frames insulin’s distributional struggles as an anomaly in an otherwise effective mechanism, RPE economists believe that corporate efforts to monopolize production and squeeze the largest possible profits out of every industry are endemic to the capitalist system.

The paradigm believes that those with diabetes should have a right to access medicine that keeps them alive. This is made possible by the fact that insulin is neither a recent nor scarce good. RPE theory would argue on a normative basis that insulin should be made freely available for those who need it and that the nature of market economics under capitalism disregards the lives of those without the income to pay for insulin prices inflated high beyond their production cost.

Originally theorized by Adam Smith, the labor theory of value (LToV) asserts that the value of a commodity originates in the amount of labor time spent during production. After being expanded upon by the original RPE theorist, Karl Marx, the LToV can be used to express how firms exploit workers and consumers in order to maximize profits. The basis of the LToV claims that the value of a commodity is a result of the average socially necessary labor time, or the time needed for an averagely skilled worker equipped with averagely efficient capital in order to produce that commodity. The capital used in production is accounted for as a summation of the "dead" labor that went into producing it. While the price of the commodity may be affected by a number of factors from scarcity, popular perception, and market power, the LToV argues that the primary contributing factor to the difference between the prices of two commodities is the amount of value imbued in that commodity during production. In the context of a perfectly competitive market, firms are price takers and consequently the value of the commodity and its market price are very similar, otherwise firms would be outcompeted. RPE theory points out how this has historically not been the case. Firms, especially those in oligopolies and monopolies, are very often able to set prices that are significantly higher than the commodity's value.

Market power creates a disconnect between a commodity's value and price. This discrepancy between production costs (in terms of value, not remuneration) and sale price is denoted as surplus and is typically expropriated by the owner of the means of production (the land and capital used) (Marx, 1867). The rate of exploitation per commodity can be calculated with the formula:

$$\text{Rate of exploitation} = \text{surplus value} / \text{total amount of wages paid}$$

For the following calculations, profit will serve as surplus value. This is in line with Marx's (1894) claim in the third volume of Capital that "Profit is the form of appearance of surplus value, which is necessarily obscured in the sphere of circulation", or in other words, the idea that profit is the numerical expression of surplus value disguised as return on capital investment rather than exploitation of labor. Without access to a company's recordkeeping, it is difficult to calculate exactly the total amount of wages paid. As a proxy, the median wage will be used for determining the degree of exploitation.

Data for Eli and Lilly  (the largest producer of insulin), shows that in 2025 the company generated a gross profit of $54.127 billion (Eli and Lilly  Company, 2025). Eli and Lilly employs 50,000 employees with a median compensation of $130,000 (Eli and Lilly  Company, 2025; Comparably, 2025). By multiplying these figures together, the company is estimated to spend approximately $6.5 billion in remuneration for the labor of its employees. Using this data, the company's rate of exploitation is approximated at 8.3272.

$$\text{Rate of exploitation}_{\text{Eli and Lilly}} = \$54.127 \text{ billion} / (\$130{,}000 * 50{,}000)$$

This figure reveals that (at median) for every 1 hour of labor a worker expends for their own remuneration, that worker is providing 8 hours and 20 minutes of labor to the capitalists running the company.

The same calculations for the second and third leading insulin giants, Novo Nordisk and Sanofi, reveal rates of exploitation of 27.77 for Novo Nordisk and 3.28 respectively (Yahoo Finance, 2025; Comparably, 2025; Novo Nordisk, 2025; Comparably, 2025; Macrotrends 2024).

$$\text{Rate of exploitation}_{\text{Novo Nordisk}} = \$250.28 \text{ billion} / (\$129{,}593 * 69{,}505)$$

$$\text{Rate of exploitation}_{\text{Sanofi}} = \$33.67 \text{ billion} / (\$123{,}497 * 82{,}878)$$

As a point of comparison, Nike has a rate of exploitation of 2.31, meaning that at the industry's lowest, the rate of exploitation is approximately 1.5 times that of the shoe producer (Nike, 2025; Comparably, 2025; Macrotrends, 2024). Part of this is like a result of Nike having competition in its market, whereas these three companies largely dominate the insulin market.

$$\text{Rate of exploitation}_{\text{Nike}} = \$19.79 \text{ billion} / (\$109{,}730 * 77{,}800)$$

Worth noting is that these companies do not solely produce insulin. Novo Nordisk's disproportionately high rate of exploitation is likely a result of its sales of Ozempic, a weight loss drug that has been making the company money hand over fist since it became more available (Synapse, 2025). As a result, the rate of exploitation does not exactly mirror the insulin market. Without access to detailed company data, it must be emphasized that these figures are rough approximations that prove the general idea that the insulin market is one that grossly exploits the workers and consumers.

While having the state intervene in order to provide insulin at the cost of production would likely be the best solution given the life saving nature of insulin's consumption, it is not feasible to consider this as an option given the current state of pharmaceutical production in the

United States. Entrenched, wealthy, and powerful interests will exert their political power to prevent such a rearrangement from being practically implemented.

As such, a reformist method of making insulin more accessible without having access to state or capital power is by reducing the information barrier in the market. Different pharmacies price the drug at different levels, meaning that some consumers are paying more than they have to (or can afford) simply because they are not buying from the cheapest seller. While this is made difficult by complications from insurance plans, the general idea of creating information for where consumers can find the lowest cost insulin can help bridge the gap between supply and demand.

Data and Evidence

**Qualitative Data**

Insulin is a large market in the United States, with over 9.3 million people who rely on some form of insulin to manage their health (Crozier-Fitzgerald, 2023, p. 97). Despite the manufacturing process for insulin being well established for decades, in the US "Between 2002 and 2013 the cost of insulin tripled, then doubled in 2016, and rose again by about 5% in 2019", which has put financial strain on the millions of people who require insulin on a regular basis. Meanwhile, research shows that a single vial of Humalog (a short-acting insulin) costs about $3 to produce, while its price to consumers is $275 (Crozier-Fitzgerald, 2023, p. 106).

In recent years the only significant decrease in consumer insulin prices has been as a direct result of government policy. The Biden administration placed a $35 per month cap on out of pocket insulin prices for those covered by Medicare and Medicaid in 2023 (Crozier-Fitzgerald, 2023, p. 109-110). Following this executive order, the big three insulin producers (Eli Lilly, Novo Nordisk, and Sanofi Aventis) each followed suit by reducing the consumer price of select insulin products. Eli Lilly was the first of the three to announce price cuts, advertising a drop from $274.70 to $66.40 per vial for their most prescribed insulin known as Humalog which is short-acting (Alltucker, 2023). Novo Nordisk followed suit soon after, cutting the price of its most sold insulin, also short-acting, known as NovoLog, from $289.36 to $72.34 per vial (Alltucker, 2023). Sanofi Aventis was the last of the three to announce price cuts in January of 2024 stating that "it will cut the price of Lantus by 78% and short-acting Apidra by 70%" (Alltucker, 2023). Sanofi has also promised that no uninsured patients will pay more than $35 for a 30 day supply of Lantus and privately insured patients will have an out of pocket cap at $15 (Alltucker, 2023).

While Sanofi's prices before and after cuts were not released, due to market behavior they were likely competitive with both Eli Lilly and Novo Nordisk before and after consumer price cuts. Even with these cuts however short-acting insulin is being sold in the $60-80 range per vial, which is still significantly more than the cost of production by a factor of 20 or more that consumers have to pay in order to stay alive in many cases. This is a common issue across the prescription drug market, as research has shown that "For every $100 spent on retail drugs, $41 go to parties in the distribution chain: wholesalers, pharmacies, pharmacy benefit managers, and insurers"(Herman, 2021, para. 29).

Companies have made these price decisions willingly and due to public scrutiny but under no legal or economic threat aside from the pressure to comply with market behavior that Sanofi faced once Eli Lilly and Novo Nordisk lowered prices. This means that at any time these companies can choose to raise prices again for their own economic gain because while public opinion will likely not be in their favor, insulin demand is inelastic and with only 3 major players the oligopoly has a lot of market power. You can see in Figure 1 how significant the trend of increasing insulin prices has been over the last 10-20 years, almost tripling since 2010, while $100 or approximately the cost of a 30 day supply of insulin in 2010, only increased to $142.33 by 2024 per inflation (U.S. Bureau of Labor Statistics). This indicates that inflation is not to blame for the drastic increase in prices, pointing back to corporate greed and a failure in the market.

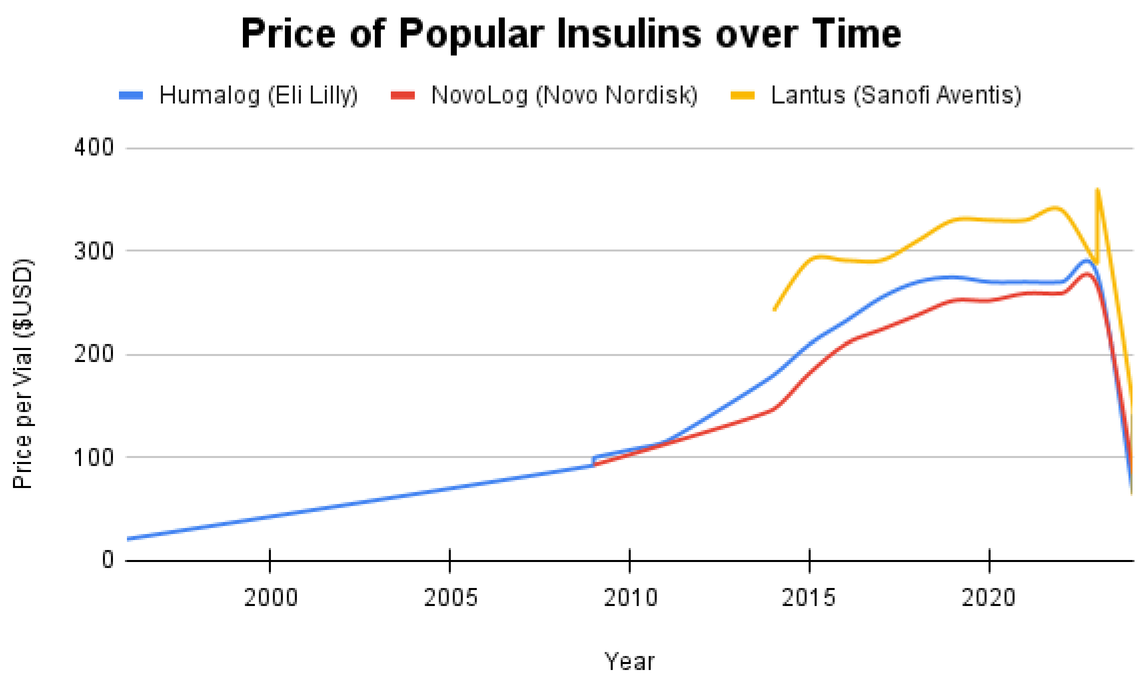


**Figure 1.** Data compiled from 3 sources (Sable-Smith, 2019), (Marsh, 2025), and (Gillett, 2019). This shows the increase in insulin prices for each of the three major insulin brands since its initial release in 1996. Data represents each company’s best selling insulin product. Lantus is a long acting insulin while Humalog and NovoLog are both short acting, however, all prices are for an estimated 30 day supply, and were selected due to the higher availability of price data over time.

**Quantitative Data**

In the search for information on prices and money flow, government sources such as the CDC and NIH as well as studies regarding the price changes and development of different insulin products over the last 30 years by other researchers and news sources with extensive resources were utilized. The primary focus of this project was on the rate of consumer price increases compared to cost of production, while controlled for inflation. Additionally, we analyzed to the best of our abilities the information available from publicly available financial records of the three major insulin producing companies. All of this data was intended to aid in answering the question of why insulin prices are so high, where the revenue is going, and the impact of these prices on the insulin market for producers, but primarily consumers.

Most research on insulin prices, while primarily numbers based, has a Radical Political Economy (RPE) leaning, suggesting motives of capital greed among the three major producers. This is why we also utilize theory analysis of both the RPE and neoclassical perspectives alongside our data, as arguments can be made using both lenses. The financial reports on the other hand, are raw numbers and have been audited by third parties in accordance with US tax regulation since all three of these companies are publicly traded. These reports, unfortunately, have restricted usefulness when it comes to determining where insulin revenue goes as R&D and production costs are not always specific and it is hard to separate insulin costs and revenue from other pharmaceuticals when these companies produce numerous drugs and treatments for ailments other than diabetes. It is, however, likely that there are minimal R&D investments into insulin at these companies considering insulin technology is and has been well established for decades.

The largest hurdle in our ability to analyze insulin prices was the inconsistency in pricing and publicly available price information. It is common that prescription drugs are  cited at different prices based on the consumer's insurance company, coverage, and even physical pharmacy. From a research standpoint that meant focusing on sales directly from these big three companies, whether they be directly to patients or to pharmacies to attempt to eliminate the additional complexity of profit distribution based on insurance coverage and pharmacy pricing. While more accurate to what these companies are directly making off of their insulin products this does also mean that our data may not reflect the everyday consumer's experience when it comes to the price tag on their insulin.

## Community Contribution

The United States government has implemented policies for limiting some types of insulin price gouging, however, the scope of these policies and their accessibility are restricted. In 2023, as part of the Inflation Reduction Act, the Biden administration enacted policy that "caps insulin out-of-pocket spending at $35 per month's supply of each insulin product covered under a Medicare Part D plan, with similar limits for out-of-pocket costs for insulin supplied under Part B, and reduces out-of-pocket drug spending in Medicare in other ways" (Sayed et al., 2023). This legislation served as an expansion of Trump's "Part D Senior Savings Model" with the primary point of difference being the accessibility of the $35 copay cap (Cubanski and Neuman, 2024).

In a similar vein to the Inflation Reduction Act, 29 states have copay legislation for capping the price of insulin with costs ranging from $25 - $100 for a month's supply of insulin for “state-regulated commercial health insurance plans” (Cubanski and Neuman, 2024). Despite this expanded availability, this legislation leaves out Americans without insurance plans.

An alternative program for alleviating overbearing insulin prices in the United States, TrumpRX, was rolled out in early 2026. TrumpRX, inspired by the website GoodRX, takes a different approach to providing affordable pharmaceutical products by instead providing coupons that consumers can use when picking up their prescriptions. Contrary to the limitations based on insurance and medicaid status present in the aforementioned policy, TrumpRX is available to all consumers of insulin, even those without an insurance plan. These coupons are negotiated by the executive and which do not formally legislate reductions in prices (Bonavitacola, 2026). In effect, this program shifts the responsibility for lowering prices from the producer to the consumer.

These programs are separated by who the burden of lowering costs is placed on and for whom they are accessible, meaning that they each have different benefits to different demographics. The copay cap model is effective in that by capping insulin prices, all those covered by the program will benefit. Furthermore, insulin only having one use case and one demographic who has a use value for the drug means that alternative groupings who gain less marginal utility from consumption will not limit the supply and drive up prices for those with greater need (Schmidtz, 2015).

Despite the success of the Inflation Reduction Act policy in reducing insulin prices for many Americans, weaknesses of the copay cap model emerge from both neoclassical and RPE perspectives. Both perspectives would note that this policy is not federally rolled out to all those who need insulin, meaning that the closing of the gap between supply and demand is not present for all demographics. Since the Inflation Reduction Act policy only covers Americans who are enrolled in Medicare, large amounts of Americans are not aided by the policy. Similarly, the copay policies by state construct a patchwork solution to the problem that leaves many Americans to slip through the gaps. In effect, the Inflation Reduction Act federally limits the supply of cheap insulin to the subsection of Americans over the age of 65, a group which by this time in their life is likely to have a larger amount of financial resources (especially given that the "baby boomer" age cohort is significantly wealthier than younger groups due to the macroeconomic conditions of the United States at the time) available to them to pay for treatments compared to younger Americans.

From a neoclassical point of view, the policy does not cure any of the underlying causes of the high prices of insulin in the first place. Putting a price cap on insulin does not fix any of the conditions that caused the discrepancy between supply and demand in the first place. Neither the free market assumptions of numerous competitive small firms nor low barriers to entry were addressed by the Inflation Reduction act and imperfect information remains ever present. Additionally, enacting "a price cap could lead to an increase in insurance premiums if the insurance company is responsible for absorbing the excess costs beyond the price cap" (Crozier-Fitzgerald, 2023, p. 110). From a neoclassical perspective, this is an ineffective use of government action that inflates the role of government in the economy for the future.

Adding to the aforementioned critiques of the demographic effects of the Inflation Reduction Act, the RPE perspective would highlight the class dynamic present in the policy's implementation. Given that baby boomers disproportionately as a generation on average possess more wealth than other generations (in part because of the macroeconomic state of the country when they were alive and in part because they've had more time to accumulate wealth), outside of cases of baby boomers who were unfortunate in their life outcomes, the generation benefits less from reduced insulin prices because of their additional ability to pay compared to the population of diabetics as a whole. In a government primarily composed of baby boomers (save President Biden who is a member of the Silent Generation), legislation to reduce their costs can be viewed as self-serving. From an RPE perspective, price caps or similar policy to lower the cost of insulin should be available to all who need such medicine.

Another critique of the model is that it is reliant on the use of state power in order to tame naked capitalist self interest. While this is by all means a beneficial use of the state, Marxist economists might point to the overwhelming power of the economic base relative to the superstructure when evaluating the longevity of such a policy. Given that the forces of capital bear immense economic and political power in the United States, reforms within the current bourgeois state are fragile and subject to being rolled back or revoked should the capitalist class need to intensify exploitation (Luxemburg, 1900).

The coupon model exemplified by TrumpRX is also subject to similar criticism from both a neoclassical and RPE perspective. Similarly to President Biden's model, from a neoclassical perspective TrumpRX does not fix any of the underlying market issues in terms of oligopoly or ease of entrance to the market. On an overarching level, TrumpRX exacerbates the example of a lack of perfect information and rationality as many individuals likely would not know about the discounts provided by TrumpRX or might willingly not use it in boycott of the current administration. In this respect, a weakness of the policy is its opt-in nature that requires additional research compared to the universal benefits provided by the Inflation Reduction Act.

The RPE perspective would criticize the TrumpRX model in an identical fashion to the Inflation Reduction Act in terms of longevity and the use of state power. With TrumpRX, due to the voluntary nature of the coupon agreements instead of a legislated price reduction, the program's fragility is increasingly apparent with "the biggest limitation of many of the steps the Trump administration is taking to lower prescription drug prices [being] the reliance on executive actions and handshake deals that may not stand the test of time" (Bonavitacola, 2026). Ironically, in terms of accessibility to the masses, TrumpRX would offer lower prices to a larger proportion of the diabetics. The actual effectiveness of TrumpRX as a government policy in this respect is variable based on the drug. While insulin lispro is listed at a lower price (especially in bulk) compared to searching on other platforms such as GoodRX or Cost Plus drugs, for a significant number of drugs listed the TrumpRX price is greater than those provided by the by non-governmental sources indicating that corporate power is not being brought to kneel as much as would be implied (Goldman, 2026). Insulin lispro is also the only type of insulin listed in TrumpRX meaning the program is entirely unhelpful for diabetics using a different type or use method.

From a neoclassical perspective, a viable solution would be the use of anti-trust measures by the federal government to break up the current oligopoly in insulin production. By having more firms competing in the insulin market, prices would be forced to go down. An RPE perspective would criticize this solution for being unrealistic since the state often looks out for the interests of the capitalist class or in other words, "The executive of the modern state is but a committee for managing the common affairs of the whole bourgeoisie" (Marx and Engels, 1848).

As mentioned in the theory section, having the government produce insulin and sell it and other medicines at a price close to the cost of production would minimize the discrepancy between supply and demand as much as possible. Alternatively, countries with universal healthcare provide lifesaving medicines like insulin to citizens for no cost. In response, neoclassical economists would argue that the government is an inherently inefficient producer and would not be able to lower costs significantly.

Non-governmental organizations can also serve to help alleviate the issue. While local, small scale insulin production would allow the breaking of this oligopoly, it is infeasible because of the complexity of the insulin production process and the capital required to do so. Instead of participating in production, non-governmental organizations can instead be useful in distribution. GoodRX and cost-plus drugs have both served as helpful options for helping consumers find coupons and low prices. Likewise, our community deliverable aims to help consumers find the best option for finding their prescription.

## Policy

The community deliverable associated with this project is a website which will provide information about the price of insulin from various companies. The website will provide this information to people in need without requiring them to see a doctor, allowing them to find information about their own health on their own terms. It will also provide accurate per-vial pricing. This community deliverable will provide more information and change the situation for many Americans with diabetes.

This community deliverable has the benefit of being cost effective. Hosting the website would not be expensive in comparison to other solutions, such as creating a new government-owned insulin production facility. Unlike GoodRx, this website would not be built with coupons or the sale of specific products in mind: its only goal would be to provide consumers with information about the products themselves so they can make the best decision for them. The hosting costs would be covered by a nonprofit organization to provide unbiased information on Insulin prices. Also, additional resources and donations would help track and monitor the different types and prices of Insulin without being influenced by revenue generated by a for-profit company.

The benefits would be massive for this cost as well, simply because consumers would have much more information about what they end up buying. To be able to easily access information about more inexpensive insulin in comparison to other sources. Insulin has very low elasticity because it is essential for many people to lead normal lives. Due to this low elasticity, people will buy it regardless of higher prices. Therefore, because people will buy insulin no matter what, having access to information pointing to the cheapest insulin will give them a real way to reduce the amount of money they spend every month without lowering their standard of living. In fact, due to the amount of money that they're saving on Insulin, they can spend that money instead on other things. During the current cost-of-living crisis, this would be very useful for many Americans who are currently overwhelmed with inflation and stagnant wages.

There comes the problem of people who can't afford enough insulin to use it to the doctor-recommended dose every month. More than 15% of adults with Insulin can't afford enough of it monthly, instead being forced to ration their insulin over a set amount of time (Gaffney, Himmelstein, et al). By providing information about the cheapest source of insulin through our community deliverables, they would, in theory, be able to ration less, as finding the cheapest source easily would allow them to purchase more. This is incredibly important due to the fact that Rationing Insulin leads to complications, most notably Diabetic Ketoacidosis. Diabetic Ketoacidosis is incredibly severe and can require hospitalization, and if you can't afford

Insulin on a regular basis, it would also be incredibly difficult to afford a visit to the Emergency Room. While many of these people's reasons for being unable to afford regular insulin are wholly unrelated to the price of insulin itself, finding them the cheapest alternative for this product will, in turn, help them afford more of it, limiting or even eliminating their rationing altogether.

Another benefit is that it could shed light on new insulin sources, potentially making the market less concentrated. These include MannKind, which produces a fast-acting, potent insulin meant to be inhaled with every meal (FDA, 2014) . With information about the different types of insulin becoming public, smaller companies could gain exposure to a larger audience, giving them more opportunities to increase their market share by encouraging people to buy more of their products. If these smaller companies can either produce different types of insulin to cater to different consumer groups, or cheaper versions of insulin to cater to people with budgetary concerns. This hopefully would also cause companies to lower prices due to increased competition. More importantly, however, it would also provide information about a new type of Insulin: Biosimilars.

Biosimilars are a new type of medication, specifically for organic compounds like insulin. Regular insulin is a biological compound that is incredibly difficult to make in a lab, in the exact same way it is made in a plant or animal cell (Heinemann, Davies, Home, et al., 2023). However, Biosimilars are highly similar compounds, made much more cheaply and tested rigorously to ensure they're as effective as the biological compounds (Ibid, 2023). This is incredibly new and interesting, since it both creates a much cheaper type of insulin and one that many people don't know about. Our website could provide information about these biosimilars, making a real difference in how much people with diabetes spend on insulin each month. This means that by providing more information about these new types of medication and competitors, it could perhaps address its most important problem: its ability to change the market.

One of the major problems with this project is how much providing insulin information will actually change the situation on the ground regarding insulin prices. While the prices of insulin from different companies are useful information to share in the market, it's almost impossible to use this information to change the reality of the Insulin market. As mentioned previously, 3 companies control around 90% of the insulin market. Therefore, our website would have a limited impact. Think of it similarly to having an app for gas prices in the nearby area. While providing information in a market with many gas stations would be more useful, the fewer the stations, the less useful the app will be. This means that, effectively, without a larger pool of companies and price points to draw from, it will be very hard for this website to have a massive impact. However, it would still have a significant impact if a company decided to make its products cheaper to beat its competitors.

This is the only truly negative situation that could result from our product. For example, if Eli and Lilly  sold cheaper but lower-quality insulin than Novo Nordisk or Sanofi, people could be enticed to buy it from them because our website says Eli and Lilly  sell the cheapest insulin. While, of course, we would attempt to provide information about the type and quality of the insulin, this may cause people to flock to this source rather than other sources, creating a Walmart-esque problem. If one company can sell more insulin at a lower price than other companies, while producing larger quantities to offset its losses, it could drive other companies out of business. This would be very problematic, since the high construction and operational overhead of an insulin factory, along with the almost 2 months required to produce insulin, would make it difficult to enter the market (Denalt, Coquet, et al.).

This project, however, could be extended in two entirely different yet both seemingly interesting ways. Firstly, and quite simply, it would show different healthcare plans and their effects under different insulin prices. This would allow the average person to truly know whether their health insurance covers the insulin they seek to buy from different sources. This would also allow for a greater understanding, as often, more expensive drugs are covered by insurance and allow people to get them for less money. While incredibly complicated due to the vast number of companies and their different plans, with a proper expansion, it would be able to help people with these different plans understand which insulin is best for them.

Secondly, if through an expansion, it would also include other medications. This app could therefore be similar to GoodRx, but with a key difference: instead of simply providing coupons for prescriptions, it could help patients find comparable drugs at different prices to discuss with their doctor, similar to Insulin. For example, ADHD drugs have wildly different pricing, and with knowledge of the difference between them in both effects and price. Consumers could make better decisions about which medications to get prescribed to them from these different sources for their daily lives. With the prior information about different medications, this website could strive to be a neutral source of information about different types of drugs for consumers at large. Therefore, this deliverable could be expanded beyond just Insulin to include other medications, providing information and helping people find the cheapest options in our current worldwide cost-of-living crisis.

All in all, our community deliverable is a cost-effective way to provide a tangible benefit to people with Diabetes across the country. While it does have problems with its ability to meaningfully change the situation, at the very least, if market conditions change and new opportunities arise from both new companies and developments, it could create a positive force that helps underdog companies and people who can’t afford insulin.

## Conclusion

This project has asserted that a failure in perfect information has occurred in the insulin market. This failure has created a need for information attempting to correct the market failure that has occurred. Imperfect information to the consumers of insulin has allowed for insulin companies and the pharmacies that sell insulin to push unfair prices onto consumers, who don’t have the information to know that they could get their medication from elsewhere, cheaper. That is why this project has ended in the creation of a website to communicate price differences between pharmacies and producers of insulin. Through the distribution of the website, this project aims to lessen the gap of information between the producers and consumers of insulin.

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